

recall at 1,000:
0.90

100,000,000 items

retrieval: precomputed vectors,
nearest-neighbour index
 $\sim 0.1 \mu\text{s}$ per item; $\approx 5 \text{ ms}$ per request

recall at 20:
0.95

1,000 candidates

ranking: cross-feature model,
user and item together
 $\sim 10 \mu\text{s}$ per item; $\approx 10 \text{ ms}$ per request

ordering:
the metric

20 shown

re-ranking: pairwise model,
rules, diversity
 $\sim 0.5 \text{ ms}$ per item; $\approx 10 \text{ ms}$ per request

end-to-end recall $\approx 0.90 \times 0.95 = 0.86$; time $\approx 25 \text{ ms}$ of a 50-ms budget